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INDUSTRY ARTICLE

A step towards digital transformation within a utility company through process automation for a sustainable energy transition

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Abstract

Sustainable energy management systems are part of the global energy transition by digitally transforming utility companies throughout the past decade. Industry 4.0 (4IR) has been adopted to improve competitiveness and align its processes with trends that require changes in the producer-consumer relationship. This study examines how AES Colombia uses Industry 4.0 to improve efficiency and promote a sustainable energy transition. It focuses on automating key processes within the company's hydropower operations, representing 6% of Colombia's energy market. The research shows that robotic-process-automation technologies reduce pricing simulation time from 27 to 5 h and tax declaration manual labour from 1560 to 584 h. These advancements enhance operational efficiency and significantly improve decision-making capabilities, thereby boosting the reliability of information utilized across various operational domains, including financial, market, and regulatory processes. The findings underscore the critical role of flexibility, cultural change, and continuous training in facilitating successful digital transformation initiatives. This work contributes to understanding how intelligent systems can optimize energy companies' processes while promoting a human-centred approach to Industry 4.0 technologies. The results underscore the importance of aligning technological advancements with sustainability goals to drive significant improvements in the energy sector, paving the way for a more resilient and adaptable energy landscape.

1 | INTRODUCTION

Nowadays, sustainable and smart energy systems are required to keep up with the pace of technological change [1] and the use and appropriation of digital technologies to enhance the relationship between producers and consumers in complex markets where renewable sources are gaining space within traditional generation systems [2]. The increase in world population [3] and the corresponding growth in energy demand have accelerated the implementation of strategies to support such digitalization of the energy sector to contribute to the fulfilment of the Sustainable Development Agenda 2030 [4], in particular with the Sustainable Development Goals (SDG): 7 'Affordable and clean energy' and 9 'Build resilient infrastructure, promote sustainable industrialization, and foster innovation'. Consequently, energy companies require modernization and adapting effectiveness, two main reasons for using this technology today [5]. Nazarov et al. [6] showed how new digital technologies have contributed to the modernization and transformation of the tax administration. Met et al. [7] showed how artificial intelligence (AI) and robotic process automation (RPA) have influenced the operations of the financial sector and their respective transformation in terms of customer service and internal operations. Huang

Abbreviations: AI, artificial intelligence; API, application programming interface; CND, Colombia's Energy National Dispatch Centre (Centro Nacional de Despacho); COP, Colombian Pesos; DIAN, Colombia's National Tax and Customs Directorate (Dirección de Impuestos y Aduanas Nacionales); DNP, Colombia's National Planning Department (Departamento Nacional de Planeación); DT, digital transformation; ERP, enterprise resource planning system; ICTs, information and communication technologies; IoT, Internet of Things (IoT); KPI, key performance indicator; ML, machine learning; MW, mega watt; MWp, mega watt peak; NCRES, non-conventional renewable energy sources; NRM, non-regulated market; RPA, robotic process automation; SDG, Sustainable Development Goal; UPME, Colombia's Mining-Energy Planning Unit (Unidad de Planeación Minero-Energética).

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and Vasarhelyi [8] applied RPA to audit processes, showing that with this modern tool, auditors do not spend time on repetitive tasks but instead focus on the essential elements that require their judgment. Kokina and Blanchette [9] showed how RPA can be used in rules-based accounting and financial processes using software bots. It is possible to find very recent studies on the digitization and automation of financial processes, such as those related to taxes in various parts of the world, as reported in Refs. [10–15].

The National Planning Department (DNP) of Colombia released two reports with detailed analyses of energy demand [16] and supply [17] within the framework of the Green Growth Mission. Regarding demand, the DNP presented the study of consumer sectors, analysed public policies, described technologies and schemes for more efficient consumption, analysed technical requirements for new technologies, and envisioned energy scenarios with updated technologies [18]. Regarding supply, it focused on non-conventional renewable energy sources (NCRES) and presented the global and national context for the energy matrix with studies carried out in Colombia; the report showed the barriers for renewables, analysed the life cycle and carbon footprint of the energy matrix, showed technical requirements for implementation and described policies, and analysed penetration scenarios with a horizon to 2040. Aligned with these DNP studies, The Mining-Energy Planning Unit presented Colombia's National Energy Plan [19], whose foundations are defined as follows: 'economic development, adaptation to climate change, and use of innovation, always reliably supplying energy demand'. With these foundations, this plan established the need to reduce the technological gap in the context of innovation, with a determined commitment to the digitization of the energy sector.

Among electricity generating companies in Colombia, there are already registered in the literature bets on the extension of the useful life of reservoirs for energy sustainability [20], renewal of power plants with technological development in infrastructure based on research in sediments [21], and digital transformation (DT) processes with the implementation of modern platforms for business management that are based on specialized tools from Industry 4.0 (4IR) [22]. Regarding the automation of financial and management processes in the regional environment, Faúndez-Ugalde et al. [23] analysed the use of AI by tax authorities in various Latin American countries, including Colombia. García et al. [24] designed and implemented a blockchain-based software tool for electronic invoicing, considering the guidelines established by the National Tax and Customs Directorate. When considering DT, it is essential to consider the effects of automation. As studied by Díaz-Pavez and Martínez-Zarzoso [25], the rapid acceleration of robot adoption can significantly impact the labour market in emerging countries. Therefore, fostering cultural change is crucial for successfully navigating these processes.

This paper presents a comprehensive case study on the successful development and implementation of intelligent automation systems across various processes within AES Colombia as part of its DT strategy. The novelty of this work lies in demonstrating the tangible impacts of 4IR tools on improving operational efficiency, reducing manual workloads, and enhancing decision-making in a utility company managing a significant portion of Colombia's energy market. These results build upon the platform implementation reported by Giraldo et al. [22] and extend beyond traditional automation by aligning process improvements with sustainability objectives. The work also addresses the critical need to align corporate practices with global and national energy transition policies accelerated by the COVID-19 pandemic [26]. At national and regional levels, this case serves as an innovative example of how a hydropower-based energy matrix [27] can leverage 4IR tools and frameworks to advance the sustainable energy transition, a priority outlined by the Inter-American Development Bank [28] and other institutions around the world [29, 30]. Furthermore, this work fulfils SDG 7 and 9 targets by supporting sustainable energy management and fostering industrial innovation. Additionally, the collaboration between industry and academia showcased in this case underscores the role of applied research in driving the adoption of cutting-edge technologies in real-world settings. Additionally, the collaboration between industry and academia showcased in this case underscores the role of applied research in driving the adoption of cutting-edge technologies in real-world settings. This partnership bridges the gap between theoretical advancements and practical applications, offering a blueprint for how 4IR innovations can be deployed to modernize and transform the energy sector, thus contributing to a sustainable energy transition. The paper is organized as follows: Section 2 describes the Colombian energy market; then, some of AES Colombia's process automation projects are addressed. Section 3 shows the benefits of the new implementations within the DT platform. Section 4 contains the discussion, and finally, conclusions are presented in Section 5.

2 | AES COLOMBIA ENERGY DIGITALIZATION PROCESS

This section outlines the key strategies and technologies implemented as part of AES Colombia's DT, focusing on integrating 4IR tools to improve operational efficiency, accuracy, and decision-making. The methods are organized into three primary areas: market simulation scenarios, financial process automation, and customer-oriented digital platforms. Each subsection presents the tools, frameworks, and processes used to enhance the company's ability to respond to the evolving challenges of the energy market, all while supporting the transition towards more sustainable energy practices. These methods reflect the core pillars of the DT strategy, as previously highlighted, contributing to more agile and scalable operations that align with the company's and the country's sustainability goals.

2.1 | Energy market context

The Colombian energy market is complex compared to several markets worldwide; several entities and companies perform generation, transmission, distribution, and commercialization

activities within laws created to promote competition [31, 32]. Electricity generation is concentrated in hydropower plants that can be affected by many external factors [27], which can cause delays in the country's energy transition towards other renewable, non-conventional generation sources without affecting the service's stability, quality, and availability. When considering energy integration and such energy transition as dominant vectors of the transformation of the country's generation matrix, great importance must be given to the participation of the generation plants that are currently in operation. Hydroelectric and thermal power plants support the Colombian electricity mix, with a share close to 68% and 30%, respectively. Several of these hydroelectric plants are in the aging stage and still operate using obsolescent information systems, control devices, data acquisition components, and operation/management software, among others.

2.2 | Company's digital transformation

AES Corporation operates two hydropower plants that serve the Colombian interconnected system. The first one is the Chivor Hydroelectric Project, a large-scale plant with an installed capacity of 1000 MW (6% of the country's capacity), whose two stages came into operation in 1977 and 1982 [20]. The second one is the small hydroelectric plant Tunjita, which has two 9.8-MW generating units that came into operation in 2016. In addition to these existing plants, AES is developing other NCRES projects as a part of the company's strategy to add 755 MW of NCRES projects to its portfolio in the country by 2025. Such NCRES plants involve a 648 MW wind power project in the department of La Guajira, 81 MWp of solar PV already in operation, and another 26 MWp for the Colombian oil company Ecopetrol. Such a portfolio requires the company to update its operation systems to facilitate supervision and integration within the energy transition scenario defined by the company and the country.

To ensure its plants' sustainable operation and digital integration, AES Colombia has been carrying out several projects focused on securing and strengthening the stability of the service provided towards the sustainable energy transition of the company's and the country's power generation system. For this, a broad 4IR-based framework for DT has been conceived [22], allowing the integration of various company processes, such as monitoring and supervision of the operation at power generation plants, financial processes, and management systems, among others, see Figure 1. This platform has been developed to improve execution times of tasks carried out manually, improve the quality of information for decision-making, and reduce risk and uncertainty generated by disaggregation of the company's information. The particular processes for which the respective innovations were implemented are related to (i) increasing the speed and reducing the operational load of analysts in the commercial area with access to quality information from market operation simulations, (ii) reducing the risk and errors by automating financial processes and tax declarations, (iii) improving the customer's experience with a platform where they can see details of consumption and services, (iv) improving the market information management system, and (v) reducing the time of billing processes. Process automation within a company like AES requires specialized 4IR tools, such as RPA, natural language processing, machine learning, and blockchain. The DT strategy of many processes in AES and its business units in Latin America will go through the robotization of processes to migrate towards business intelligence in the energy transition scenario. The first three innovations are the centre of the results provided in this case report.

2.2.1 | Market simulation scenarios

Giraldo et al. [22] reported that AES Colombia's digital platform comprises several applications. The first automated processes, including one for the supply, allow carrying out computer simulations of multiple dispatch scenarios using mathematical logic and the variables that are updated by the National Dispatch Centre (CND) and evaluating the economic result with each simulated dispatch scenario to support the daily offer strategy for the proposals that sent to the energy market [32]. This stage of the DT has been developed to increase the platform's efficiency and improve price predictions, support, and constant updating of regulatory changes in the energy market in Colombia. This includes: (i) developing additional automatic simulation scenarios; (ii) validating prices that are above the maximum marginal price and comparing them with the actual price of the first day of the month; (iii) generating automatic messages with information to predetermined recipients; (iv) eliminating validation steps that cause the offer price of the plants to go down; and (v) obtaining an approximate actual generation plan from the optimized ideal generation.

2.2.2 | Financial processes automation

To reduce risk, errors, and execution times within financial processes, several automated processes have been implemented around tax declarations [8, 13] and contracting models. RPA is a definable, repeatable, and rule-based technology that allows the company to have process traceability and assign a robot (or bot, represented by software) that oversees executing the process [5]. The implementation of this technology for financial processes requires, among others: (i) Cultural changes in the teams so that competencies in Information and Communication Technologies are created/enhanced, and data analytics capabilities are improved to identify failures and inconsistencies in processes; in this way, people can focus efforts on processes and not on tools (many of them manual until now). (ii) Reduction of hours for information processing. (iii) Actively work on the digitization of financial processes. (iv) Continuous improvement. For instance, in the case of tax processes, this development includes the automation of extraction and consolidation of reports from the enterprise resource planning system, preparation of tax balances, and the automation and standardization of processes specified by government rules, among others (see Figure 2).

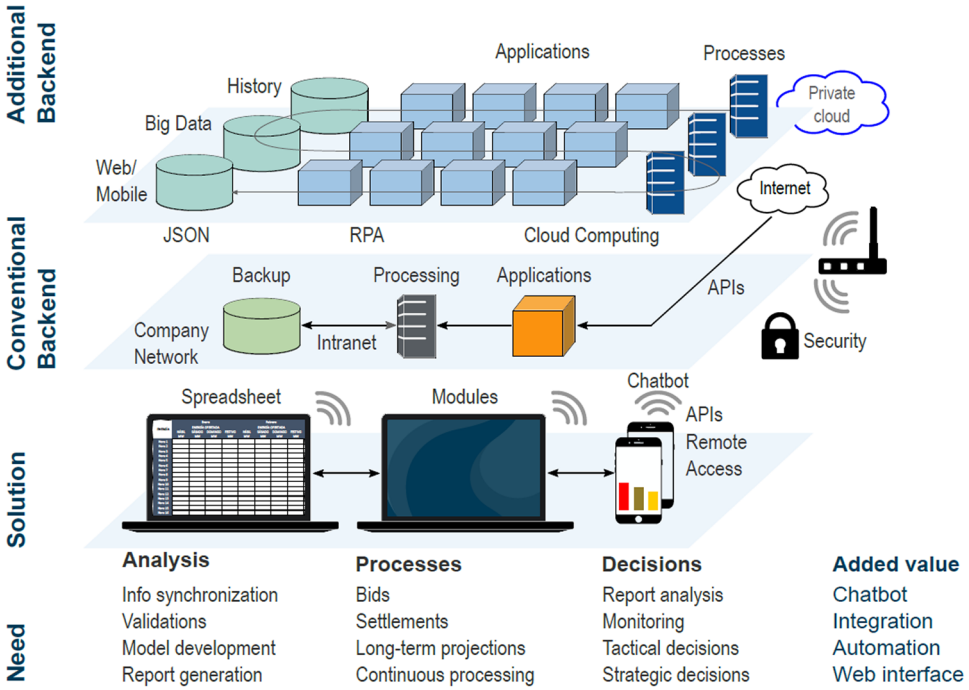


FIGURE 1 AES Colombia digital transformation (DT) architecture, reported by Giraldo et al. [22].

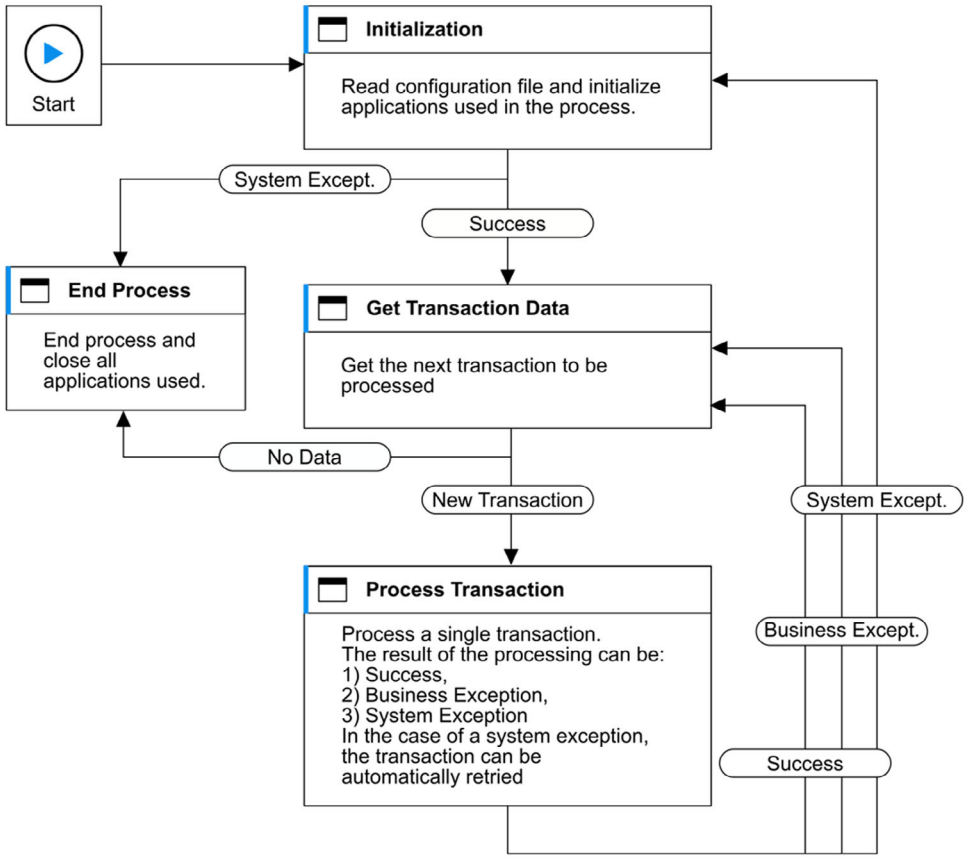


FIGURE 2 Financial processes automation.

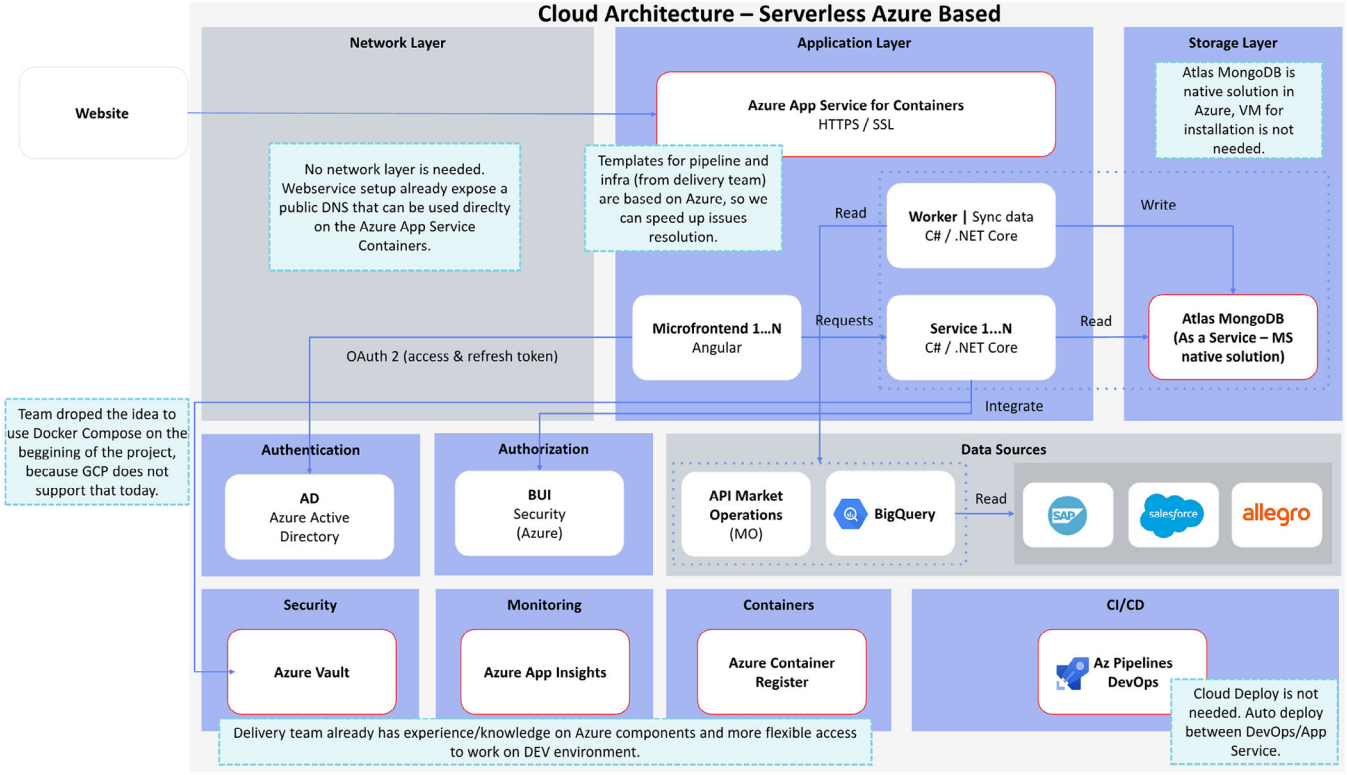


FIGURE 3 Clean energy navigator architecture.

The UML diagram represents the financial process automation workflow, which shows interactions between different system components during transaction processing. The process begins when a request is received by the application programming interface (API) Gateway, which forwards it to the Controller. The Controller then delegates the request to the service layer, which first tries to retrieve data from the database. If no data is found, the service initiates a new transaction with an external system, such as a third-party financial service. Depending on the response (success, system exception, or business exception), the service sends the result back to the Controller, which relays it to the API Gateway. The Gateway then communicates the outcome to the client, ensuring seamless automation of financial transactions with robust error handling.

2.2.3 | Customer-oriented platform

To enhance the relationship with customers [33], within the framework of the customer-focused strategy, an intelligent platform has been developed for customers to interact with AES; for instance, they access consumption and service details, representing an improved experience in the commercial relationship. The development of this platform, called clean energy navigator (CEN), seeks to improve the speed and quality of delivery of digital solutions for these stakeholders. The development team defined a software architecture, with the guidelines released by the AES digital factory, so that the intelligent platform meets the appropriate standards to be integrated into the DT strategy. Scrum methodology, an agile methodology based on incremental development, has been used to develop these software components. The developed components underwent the respective AES cybersecurity tests so that only modules, codes, or components without vulnerabilities were accepted and are being used. Figure 3 contains the architecture that has been used for the customer-oriented platform. Figure 3 illustrates a Serverless Azure-based cloud architecture designed for a utility company's DT towards sustainable energy management. The architecture integrates key components across various layers, including Azure App Service for containers for hosting microservices, Atlas MongoDB for storage, and Azure Active Directory for authentication. It leverages Azure Vault for security, Azure App Insights for monitoring, and Azure Pipelines/DevOps for continuous integration and deployment. The design emphasizes ease of use, with no need for a traditional network layer or cloud deployment, and highlights the team's prior experience with Azure, allowing for flexible development and streamlined issue resolution.

3 | RESULTS

The following sections present the outcomes of the DT initiatives undertaken within AES Colombia, highlighting the improvements brought by 4IR tools across various operational areas. These results are categorized into three key examples: enhancing market simulation scenarios, automating financial

processes using RPA, and deploying a customer-oriented platform to improve energy consumption management. Each section details how the implemented solutions have contributed to significant efficiency gains, reduced operational risks, and supported better decision-making, ultimately facilitating a more sustainable energy transition within the company.

3.1 | Example of market simulation scenarios

The platform developed to simulate market scenarios increases speed and reduces the operational burden of commercial area analysts using the best technology and information available. The modular architecture allows reducing response times, increasing the depth of analysis, and the capacity for innovation. The automatically generated reports enable the experts to analyse the results and project economic variables. These reports include information on contributions, generation by technology, shipments, purchases, and significant stock market sales. For instance, the previous work from Giraldo et al. [22] reported time savings from report generation that went down from 108 to 32 h/month. In this case, we can report the development of the long-term module that allows one to project energy prices in the short, medium, and long-term based on an energy dispatch optimization model.

At the beginning of the project, automatic analysis was limited to eight market simulation scenarios, which were insufficient when considering variations in the behaviour of other agents and competition for the AGC. Manual processing of the complementary simulations required approximately 7300 h/year. The case simulation manual represented the most significant challenge in tackling this project, which involved possible biases in the market reading and delays in defining the scenarios for optimizing the commercial strategy. These factors impacted the day-to-day supply decision and, ultimately, the income level. The impacts derived from the implemented improvements are reflected in the following aspects: Reduction in processing times: The speed of data processing represents a 92% reduction in person-hours dedicated to this task, which implies more time for value analysis that supports effective decision-making. Improved accuracy: with automatic simulations, errors associated with the assessment of the movement of some agents, as well as operational failures and human error, are omitted. Under these conditions, the accuracy of the sought-after generations is significantly increased. With the additional simulation cases, the resources that previously had to perform the calculations and estimate the variables manually now have more time for the further analyses required by the market dynamics. The strategy's objective is to achieve more significant income by minimizing risk. As the system provides greater visibility of the market variables and their interactions, mitigating the risks inherent to the process is favoured, and thus, the benefits are increased through a better income. A more significant number of generation scenarios facilitate the optimization of the commercial strategy, providing a solid basis for decision-making in the day-to-day operation and over an extended time frame. The project has undoubtedly provided technical, human, and financial benefits, becoming an alternative response to the current commercial management requirements without neglecting sustainability, scalability, and future responsiveness.

TABLE 1 Time comparison to perform a price projection study.

Report type	Time before DT (h)	Time after DT (h)
Information update	16	0.1
Definition of scenario assumptions	3	0.5
Scenario simulation	4	4
Generation of results	4	0.2
Total projection time	27	4.8

The result obtained with this development simulates the economic dispatch executed by the CND for a required time horizon, with the following output variables: projection of the behaviour of the stock market price; projection of the ideal and real generation of each plant in the system, including new projects that have been awarded in long-term auctions; contract price forecasting; and the expected expansion of the generation matrix. Additionally, the long-term module allows evaluation of the impacts of new regulations, reservoir management, contracting levels, declaration of costs, and incorporation of new plants, among others, and allows analysing the behaviour of solar and wind plants and evaluating their impact on the operation of the system. The inputs of this module include:

- Demand growth
- Occurrence of El Niño or La Niña
- Hydrology
- Production costs of thermoelectric plants
- Fuel availability
- Average fuel prices: LNG, diesel, coal, and gas
- New generation projects

Table 1 compares the time required to perform a price projection study before and after DT. To test the simulation platform, backtest simulations were conducted using data from the Colombian market's operator XM [34]; Figure 4 compares daily price simulation results to the actual value of the stock price for 2 years (2017–2018).

3.2 | Example of financial processes automation

In Colombia, AES started digitizing five tax processes using RPA. They aim to streamline these processes, eliminate manual intervention, and increase efficiency while minimizing risk. By doing so, AES Colombia has freed up resources to focus on more strategic and value-added activities, such as analysing data and identifying opportunities for tax savings. Table 2 compares the time required to perform tax processes before and after RPA implementations.

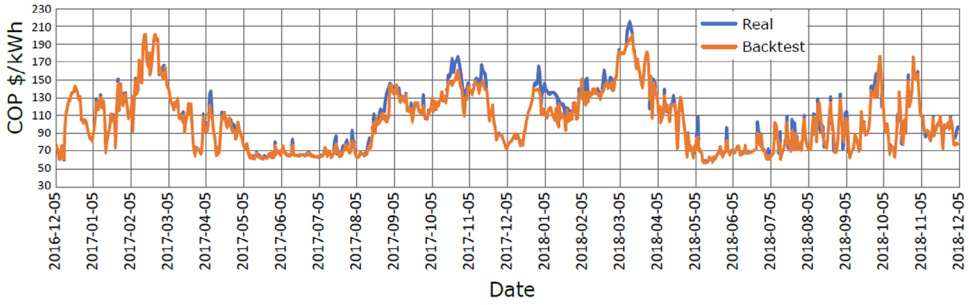


FIGURE 4 Daily price simulation (in Colombian pesos, COP) compared to the actual stock price value for 2017–2018.

TABLE 2 Time comparison for tax processes before and after robotic process automation (RPA).

Task type	Time before RPA (h/year)	Time after RPA (h/year)
Bots	–	976
Manual	1560	584

Before the implementation of RPA, handling tax processes was entirely manual. This included data entry, calculations, and compliance checks, which were time-consuming and prone to errors. On average, these manual processes require approximately 1560 h annually. The manual approach led to frequent errors in data entry and calculations, delays in processing, and difficulties in ensuring compliance with evolving tax regulations. This increased the risk of fines and penalties and diverted resources from more strategic activities.

The introduction of RPA has drastically reduced the time required for tax processes. The automation has cut the manual processing time from 1560 to 584 h per year (see Table 2). This represents a significant 62% reduction in human hours dedicated to tax compliance tasks. The automation of these processes has streamlined operations and improved overall efficiency. RPA handles repetitive tasks such as data entry and calculations, improving the accuracy of tax processes. Automation minimizes human error, ensuring that tax calculations are precise and compliant with current regulations. This reduction in errors has led to fewer penalties and a lower risk of legal consequences. Additionally, RPA has freed up valuable human resources by automating routine and time-consuming tasks. These resources can now be redirected towards more strategic activities such as analysing data, identifying tax-saving opportunities, and improving overall tax strategy. This shift enhances operational efficiency and contributes to better financial management. Moreover, RPA ensures that tax processes are carried out consistently and following the latest regulations. Automated systems can be updated more quickly to reflect changes in tax laws, reducing the risk of non-compliance and associated penalties. This improvement in compliance also enhances the company's reputation and reduces the likelihood of legal issues. In addition, RPA enabled real-time monitoring of key performance indicators (KPIs) related to tax processes. This capability provides better insights into tax compliance and performance, allowing for more informed decision-making and timely strategy adjustments. Finally, automation has led to standardizing tax processes and documentation, simplifying reporting, and ensuring consistency across different regions and departments. This standardization helps maintain uniformity and accuracy in tax filings.

General achievement results obtained with the automation of tax processes can be summarized as follows: (i) reduced time spent on manual data entry and calculations, freeing up resources for other tasks due to the human hour's reduction in tax compliance processes, (ii) online monitoring of specific KPI key for the tax area, allowing for better insights and decision-making; (iii) improvements of controls layers and compliance with tax regulations and laws, reducing the risk of legal and financial consequences; (iv) standardization of all process and working papers; (v) increased accuracy of tax calculations resulting in reduced errors and penalties, reducing risks of sanctions and tax fines; and (vi) increased efficiency in tax preparation and reporting, resulting in faster turnaround times.

3.3 | Example of the customer-oriented platform

Nowadays, the customer-oriented platform CEN is a highly analytical tool that allows the user to identify opportunities for improvement in consumption. We started by uploading billing and consumption data into the platform. We performed several tests for the first group of ten clients, which made the initial launch possible by January 2022 during a meeting with representatives from several companies. The tool allows them to (i) view consumption and cost information for each contracted site with a high level of detail, (ii) access invoices that are soon due or invoices with pending payment, (iii) compare electrical charges with past periods, (iv) view the breakdown of invoices, and (v) compare monthly usage and identify differences. The data loaded into the application allows accessing consumption (active energy), billing information, trends, and performing different analyses. The users can download data and reports in CSV, PDF, and PNG formats. The application also allows the user to compare similar industrial facilities and suggests energy savings and efficiency alternatives; hence, the users need to customize and characterize their installations as much as possible. By 2024, indicators will be generated regarding the tool's usabil-

ity (total enabled companies, active users, and adequate records, among others). The use of the CEN portal has had a direct impact on the qualification of users, given its importance in the improvement of their processes due to the reduction of time, the traceability of data, and comparative consumption analysis as a basis for the formulation of action plans for the efficient use of energy. The tool has been implemented for 94% of AES clients, with 48 users benefiting from the functionalities mentioned earlier.

4 | DISCUSSION

The use of 4IR technologies is pivotal for steering a sustainable energy transition within a company such as AES Colombia, which operates hydropower plants, with future expansion to integrate solar and wind power facilities. As the company diversifies its energy portfolio with the integration of NCRES, automation becomes even more crucial as more variables and processes will put more complexity into the company's operations. Integrating 4IR technologies enhances operational efficiency and positions the energy company at the forefront of a sustainable energy paradigm, promoting a harmonious coexistence between technological advancement and environmental responsibility. Consequently, we identified flexibility as a critical element within these DT processes, both in contractual terms and in tacit or express agreements derived from the interaction among those in charge of planning, development, and implementation of DT projects.

Promoting process innovation as part of the internal and external culture constitutes one of the organization's work pillars within the framework of its sustainability strategy. Overall, 14% of capital investments during the previous year were allocated to information technology and telecommunications projects, thereby impacting the goal of increasing research and updating industrial technologies, established in the 2030 agenda for the sustainable development goal SDG 9: industry, research, and infrastructure.

However, implementing the different solutions in this automation project, we found technical difficulties related, for instance, to the effectiveness of interface connection, generating inconsistencies, and delay time to obtain the solution. This situation implied performing reviews and reprocessing, situations that can be avoided in the future through validation tests in the early stages of the project. On the other hand, the human factor cannot be ignored, the usual way of developing the different activities about digitalization processes and the opposition towards adopting new work schemes and tools, which indicates that cultural change is a fundamental element for the success of AES's DT projects.

It is necessary to provide open test environments to the entire team during the implementation of DT. This can promote familiarity with the use of new tools coming from 4IR and make the team aware of different restrictions for the automatic operation of the process. This will facilitate interaction with the proposed change in the process before going into production to identify those aspects that need to be adjusted at the energy company. For example, regarding some of the tax returns that were automated, there is no uniformity in the returns and annexes to be submitted to each municipality in Colombia as each city is free to establish its withholding rates and the forms of presentation and payment processes; hence, we found high complexity to standardize this tax process considering Colombian regulations.

The experience gained from successful DT projects and those still in the implementation phase at AES Colombia allows one to identify lessons learned. It is necessary to have suppliers with vast operational knowledge about the object of work: the energy market. The difficulties caused by poor communication, due to the lack of understanding of a developer who may have all the credentials from a technological point of view but lacks mastery of regulatory aspects and the Colombian energy market, lead to more significant damage and block fluid developments of the projects, and in the end, prevent their practical realization, discouraging the incorporation of improvements, and in general, fostering an environment that is not favourable for change.

This case study contributes to the existing literature by providing empirical evidence of how integrating 4IR technologies can transform operational practices within energy companies like AES Colombia, facilitating their efforts towards a sustainable energy transition. Although previously published research has mainly focused on specific sectors such as tax administration and financial operations, this work expands the scope to encompass the energy sector's unique challenges and opportunities in digitization. The findings demonstrate that implementing advanced digital solutions, including RPA and simulation technologies, enhances operational efficiency and fosters a more robust relationship between energy producers and consumers. Moreover, this study highlights the importance of cultural and organizational adaptability in successfully navigating the DT process, reinforcing the notion that a supportive internal culture for sustainable outcomes must complement technological advancement. By aligning the operational advancements with Colombia's broader SDG, this research lays the groundwork for future investigations into the strategic role of DT in achieving resilience and innovation within the energy sector.

Despite the promising advancements in AES Colombia's DT, several limitations arise from this work. First, the dependency on legacy systems can hinder the full integration of 4IR technologies, as older infrastructure may not be compatible with modern digital solutions. This can lead to inefficiencies in data collection and analysis, limiting the potential benefits of automation. Additionally, the variations in local tax regulations across municipalities in Colombia pose a significant challenge for standardizing automated financial processes. The lack of uniformity in tax return submissions complicates automation efforts, requiring continuous adjustments and re-evaluation of the implemented solutions. Furthermore, resistance to change among employees and the prevailing organizational culture can impede the adoption of new technologies and processes. Although implementing open test environments can foster familiarity and ease the transition, it does not guarantee widespread acceptance or engagement from all team

members. Addressing these limitations will be crucial for optimizing the DT journey and achieving the desired outcomes in the future.

5 | CONCLUSIONS

This work addressed implementing modern 4IR -based strategies to improve execution times and the quality of information in tasks carried out manually in several processes of an energy company that supplies around 6% of the Colombian energy market. The execution of several process automation projects allowed for reducing price simulation time from 27 to 5 h and reducing the risk and errors by automating financial processes such as tax declarations, where manual work time went from 1560 to 584 h by using RPA.

In addition, the process innovations based on 4IR technologies have provided greater efficiency when making decisions and making more efficient use of information to support the electricity market of the national interconnected system of Colombia. Thanks to the increased reliability, this will impact end customers' experience in the Colombian market through automation at various points of AES's sustainable operation within the energy transition. The transition from a local process management strategy in the country to a global one has also required synergies, learning, teamwork, communication skills, and mastery of different technological tools, whose implementation leads to human-centred automation in various processes at an energy company.

As shown in this work, 4IR-associated technologies can be applied to different processes within the energy industry, offering companies technical advantages while migrating towards a sustainable energy transition. These intelligent systems can guide and support decision-making in companies and potentiate the conversion of energy into monetary value based on large amounts of information from various sources (e.g., variation in resources, consumption, and generation over time, status of equipment operation variables, economic indicators, variables from the energy market, among others). Using such quality information in various processes (operation, market, financial, tax etc.) allows energy companies to reduce the risks of transactions in market operations. However, preparing human resources for the digital transition becomes a pivotal factor as people are the centre of processes that make such quality valuable information in the transition to Society 5.0, defined as the human-centred application of 4IR.

Finally, AES Colombia's DT journey presents both opportunities and challenges. Future work will involve scaling up the automation projects across all operational areas, ensuring that lessons learned from initial implementations are systematically integrated into subsequent phases. This includes enhancing collaboration with technology partners with deep knowledge of the energy sector, thereby minimizing the communication gaps that have previously hindered project progress. Additionally, as the company seeks to integrate more NCRES, developing robust data management frameworks will be crucial to handle the increasing complexity of operations and maintain compliance with evolving regulations. Training and upskilling the workforce will also be essential, as preparing human resources for the digital transition is critical for maximizing the value of quality information in the context of Society 5.0. Addressing these challenges through ongoing investment in technology, culture change, and employee engagement will be fundamental for AES Colombia to navigate its future in a rapidly evolving energy landscape successfully.

AUTHOR CONTRIBUTIONS

Nhora Parra, Sandra Giraldo, David la Rotta, Betsy Gómez, Yissel Mejía, and Laidi Morales conceived the idea and conducted simulations, analyses, and discussions of the results. They also contributed to preparing the first draft and final manuscript. César Nieto-Londoño and Rafael E. Vásquez participated in the analysis and discussion of the results and contributed to writing and editing the first draft and the final manuscript.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

Data are available on request due to privacy/ethical restrictions.

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